Lab 4: Same brief to every subagent, same findings four times

Topic 1: Agentic Architecture & Orchestration · Task 1.2: Orchestrate multi-agent systems with coordinator-subagent patterns

Runs on Simulator 6: Structured Data Extraction · Self-practice

Source: SkillCertPro practice exam Part 1, Q36 (unofficial third-party question)

Commit to an answer first

During evaluation, this team asks Claude Code to use subagents to investigate extraction errors across 600 vendor invoices, intake forms, and contracts. The current approach gives each subagent the same instruction (find causes of schema-validation failures across all documents), producing repeated observations and a large synthesis context. What change best partitions the investigation while preserving synthesis quality?

A. Assign each subagent a distinct document group or extraction concern, provide needed context, and require concise findings for synthesis.

B. Create one nested subagent chain per document type, require verbose summaries at each level, and synthesize the final outputs.

C. Assign overlapping document subsets to multiple subagents, ask each to find any schema issue, and reconcile duplicates afterward.

D. Ask every subagent to review the full corpus independently, compare their repeated findings, and merge common schema issues.

Simulate the solution

```
# In the browser: open Simulator 6, then Lab 4
cd labs
uv run serve.py

# In Claude Code, from the labs folder
/lab 4

# In a terminal, from simulators/simulator-6-structured-extraction/
uv run extract.py --case UC-36
```

What you should see

Every document lands in exactly one group, and each subagent gets a line-capped deliverable.

evidence/cases/UC-36.json holds a partition where every document appears in exactly one group, one rendered prompt per subagent with a line-capped deliverable, and the expected answer.

Open `simulators/simulator-6-structured-extraction/evidence/` and find it yourself.

The answer

A. Give each one a distinct document group or concern, and demand concise findings.

The documented parallel-research pattern spawns subagents for independent investigations and synthesises in the main conversation. Partition along a boundary that keeps paths non-overlapping: document family, source type, schema section or failure mode. Isolation keeps exploratory work out of the main thread, but returned output is not free. Long reports still bloat synthesis, so each delegation needs context, explicit scope and a compact deliverable format.

Why the others fail

B. Nesting plus verbose summaries at every level bloats the synthesis it is meant to protect.

C. Overlap without clean boundaries means duplicated work and a reconciliation burden.

D. Redundancy rather than partitioning, which is the problem being fixed.

Check yourself

- Learner's partition has no overlapping document ids and names the concern per group.
- Learner caps the deliverable and explains why isolated context does not make returned output free.
- Learner states what context each subagent must be given because it inherits none.

Now change one thing

Change DOC-04's printed total in fixtures/invoices.txt, rerun, and watch by_document_type.invoice fall while the others hold.

Third-party practice material from SkillCertPro, reconstructed from a video transcript in the Tertiary Markdown Vault. Not a real exam item. Wording may differ from the original.